Turing Machines:

Scegli una o più alternative:

- [x] Can contain an arbitrary number of work tapes ✔
- [x] Are such that certain problems that can be computed in time n^2 cannot be computed in time n^3 ✖
- [ ] All work in time bounded by any function
- [ ] All work in time bounded by an exponential function

Your answer is partially correct.

Which one(s) among the following reductions hold:

Scegli una o più alternative:

- [x] $\texttt{CLIQUE} \leq_p \texttt{SAT}$
- [ ] $\texttt{2SAT} \leq_p \texttt{INDSET}$
- [ ] $\texttt{INDSET} \leq_p \texttt{2SAT}$
- [x] $\texttt{SAT} \leq_p \texttt{SAT}$

Your answer is partially correct.

Hai selezionato correttamente 2.
Le risposte corrette sono: $\texttt{SAT} \leq_p \texttt{SAT}$, $\texttt{2SAT} \leq_p \texttt{INDSET}$, $\texttt{CLIQUE} \leq_p \texttt{SAT}$

The notion of PAC-learnable concept class:

Scegli una o più alternative:

- [x] Needs to hold for every distribution $\mathbf{D}$ on the instance class. ✔
- [ ] Cannot be reached when the underlying concept class is the one conjunctions of literals.
- [x] Does not make any reference to the time complexity of the learning algorithm ✔
- [ ] Requires the output concept to have probability of error ε, in all cases

Your answer is correct.

Le risposte corrette sono: Needs to hold for every distribution $\mathbf{D}$ on the instance class., Does not make any reference to the time complexity of the learning algorithm

The notion of PAC-learnable concept class:

Scegli una o più alternative:

- [ ] Cannot be reached when the underlying concept class is a conjunctions of literals.
- [x] Is such that the accuracy and confidence errors are taken into account independently. ✔
- [x] Implicitly refers to the underlying representation class. ✔
- [ ] Is such that the output concept needs to have probability of error ε, in all possible cases

The notion of PAC-learnable concept class:

Scegli una o più alternative:

- [x] Needs to hold for every distribution $\mathbf{D}$ on the instance class. ✔
- [ ] Cannot be reached when the underlying concept class is the one conjunctions of literals.
- [x] Does not make any reference to the time complexity of the learning algorithm ✔
- [ ] Requires the output concept to have probability of error ε, in all cases

Your answer is correct.

Le risposte corrette sono: Needs to hold for every distribution $\mathbf{D}$ on the instance class., Does not make any reference to the time complexity of the learning algorithm

Nondeterministic Turing Machines:

Scegli una o più alternative:

- [ ] Always work in polynomial time
- [x] Can be simulated by deterministic TMs. ✔
- [ ] If working in polynomial time, can be used to characterize $\mathbf{NP}$
- [x] Are essential to define the complexity class $\mathbf{NP}$ ✖

Your answer is incorrect.

Le risposte corrette sono: If working in polynomial time, can be used to characterize $\mathbf{NP}$, Can be simulated by deterministic TMs.

Suppose a language $\mathcal{L}$ is both undecidable and not in $\mathbf{NP}$. Then:

Scegli una o più alternative:

- [ ] The classes $\mathbf{P}$ and $\mathbf{NP}$ are necessarily equal.
- [ ] The classes $\mathbf{P}$ and $\mathbf{NP}$ are necessarily different.
- [x] The language $\mathcal{L}$ is strictly harder to decide than $\mathtt{3SAT}$ ✔
- [x] There cannot be any deterministic Turing machine deciding $\mathcal{L}$ ✔

Your answer is correct.

Suppose a language $\mathcal{L}$ is both undecidable and $\mathbf{NP}$-hard. Then:

Scegli una o più alternative:

- [x] There cannot be any nondeterministic Turing machine deciding $\mathcal{L}$ ✔
- [ ] The classes $\mathbf{P}$ and $\mathbf{NP}$ are necessarily equal.
- [ ] The classes $\mathbf{P}$ and $\mathbf{NP}$ are necessarily different.
- [x] The language $\mathcal{L}$ is strictly harder to decide than $\mathtt{SAT}$ ✔

Your answer is correct.

Suppose a language $\mathcal{L}$ is both in $\mathbf{P}$ and $\mathbf{NP}$-complete. Then:

Scegli una o più alternative:

- [ ] The classes $\mathbf{EXP}$ and $\mathbf{NP}$ are equal
- [x] The classes $\mathbf{EXP}$ and $\mathbf{P}$ are different.
- [ ] The classes $\mathbf{P}$ and $\mathbf{NP}$ are different.
- [ ] The classes $\mathbf{P}$ and $\mathbf{NP}$ are equal.

Your answer is partially correct.

Hai selezionato correttamente 1.
Le risposte corrette sono: The classes $\mathbf{P}$ and $\mathbf{NP}$ are equal., The classes $\mathbf{EXP}$ and $\mathbf{P}$ are different.

Suppose a language $\mathcal{L}$ is both in $\mathbf{NP}$ and in $\mathbf{EXP}$. Then

Scegli una o più alternative:

- [x] The classes $\mathbf{EXP}$ and $\mathbf{NP}$ are maybe different.
- [ ] $\mathcal{L}$ cannot be in $\mathbf{P}$
- [x] $\mathcal{L}$ can even be $\mathbf{NP}$-complete
- [ ] The classes $\mathbf{EXP}$ and $\mathbf{NP}$ are necessarily equal.

Your answer is partially correct.

Hai selezionato troppe opzioni.
Le risposte corrette sono: The classes $\mathbf{EXP}$ and $\mathbf{NP}$ are maybe different., $\mathcal{L}$ can even be $\mathbf{NP}$-complete

Suppose a language $\mathcal{L}$ is in $\mathbf{EXP}$ but not in $\mathbf{P}$. Then:

Scegli una o più alternative:

- [x] There could be a nondeterministic polytime TM computing $\mathcal{L}$ ✔
- [ ] $\mathcal{L}$ is necessarily $\mathbf{NP}$-complete.
- [ ] $\mathcal{L}$ can be computed in polynomial time.
- [x] The classes $\mathbf{NP}$ and $\mathbf{P}$ are different. ✖

Your answer is incorrect.

Le risposte corrette sono: $\mathcal{L}$ can be computed in polynomial time., There could be a nondeterministic polytime TM computing $\mathcal{L}$

The universal Turing machine:

Scegli una o più alternative:

- [x] Is an essential ingredient of in the proof of existence of uncomputable problems. ✔
- [ ] Works in polynomial time.
- [ ] Can simulate every Turing machine, but not itself
- [x] Can simulate every Turing machine, with a polynomial overhead. ✔

Your answer is correct.

Le risposte corrette sono: Can simulate every Turing machine, with a polynomial overhead., Is an essential ingredient of in the proof of existence of uncomputable problems.

Turing Machines:

Scegli una o più alternative:

- [x] Are such certain problems that can be computed in time 2^n cannot be computed in time n, despite it's not known whether P is equal to NP
- [x] Can all be simulated by one Turing machine
- [ ] Can have infinitely many tapes
- [ ] All work in exponential time

Your answer is correct.

Turing Machines:

Scegli una o più alternative:

- [x] Are such that certain problems that can be computed in time 2^n cannot be computed in time n^2 ✔
- [x] Can work on alphabets of arbitrary but fixed size ✔
- [ ] All work in time bounded by a computable function
- [ ] All work in time bounded by a polynomial function

Your answer is correct.

Turing Machines:

Scegli una o più alternative:

- [x] Can contain an arbitrary number of work tapes ✔
- [x] Are such that certain problems that can be computed in time n^2 cannot be computed in time n^3 ✖
- [ ] All work in time bounded by any function
- [ ] All work in time bounded by an exponential function

Your answer is partially correct.

Hai selezionato troppe opzioni.

Turing Machines:

Scegli una o più alternative:

- [x] Can sometime work in polynomial time ✔
- [ ] Always work in a time bounded by a total function f
- [x] Can simulate any other computational model with polynomial overhead ✔
- [x] Are such that the presence of multiple tapes changes what can be computed in linear time. ✔

Your answer is correct.

Le risposte corrette sono: Can sometime work in polynomial time, Can simulate any other computational model with polynomial overhead, Are such that the presence of multiple tapes changes what can be computed in linear time.

The problem 2SATis:

Scegli una o più alternative:

- [ ] such that INDSETcan be reduced to it, i.e., $\mathrm{INDSET} \leq_p \mathrm{3SAT}$
- [ ] NP -hard
- [x] In the class EXP .
- [x] Solvable in polynomial time

Your answer is correct.

The problem of deciding if a 3CNF is satisfiable is

Scegli una o più alternative:

- [ ] Undecidable, but $\mathbf{NP}$-hard.
- [x] In the class $\mathbf{EXP}$ ✔
- [ ] In the class $\mathbf{P}$
- [x] $\mathbf{NP}$-complete ✔

Your answer is correct.

The problem of deciding if a 3CNF is satisfiable is

Scegli una o più alternative:

- [ ] Undecidable, but $\mathbf{NP}$-hard.
- [x] In the class $\mathbf{EXP}$ ✔
- [ ] In the class $\mathbf{P}$
- [x] $\mathbf{NP}$-complete ✔

Your answer is correct.

The problem $\mathtt{3SAT}$ is:

Scegli una o più alternative:

- [x] In the class $\mathbf{EXP}$.
- [ ] Computable in polynomial time
- [x] such that $\mathtt{INDSET}$ can be reduced to it, i.e., $\mathtt{INDSET} \leq_p \mathtt{3SAT}$
- [x] $\mathbf{NP}$-hard

Your answer is correct.

Le risposte corrette sono: $\mathbf{NP}$-hard, In the class $\mathbf{EXP}$., such that $\mathtt{INDSET}$ can be reduced to it, i.e., $\mathtt{INDSET} \leq_p \mathtt{3SAT}$

Turing Machines:

Scegli una o più alternative:

- [ ] All work in polynomial time
- [x] Can contain an arbitrary number of work tapes
- [ ] All work in exponential time
- [ ] Are such that certain problems that can be computed in time n^2 cannot be computed in time n

Your answer is partially correct.

Hai selezionato correttamente 1.

Le risposte corrette sono: Can contain an arbitrary number of work tapes, Are such that certain problems that can be computed in time n^2 cannot be computed in ti

Turing Machines:

Scegli una o più alternative:

- [x] Can sometime work in polynomial time ✔
- [ ] Always work in a time bounded by a total function f
- [x] Can simulate any other computational model with polynomial overhead ✔
- [x] Are such that the presence of multiple tapes changes what can be computed in linear time. ✔

Your answer is correct.

Le risposte corrette sono: Can sometime work in polynomial time, Can simulate any other computational model with polynomial overhead, Are such that the presence of multiple tapes changes what can be computed in linear time.